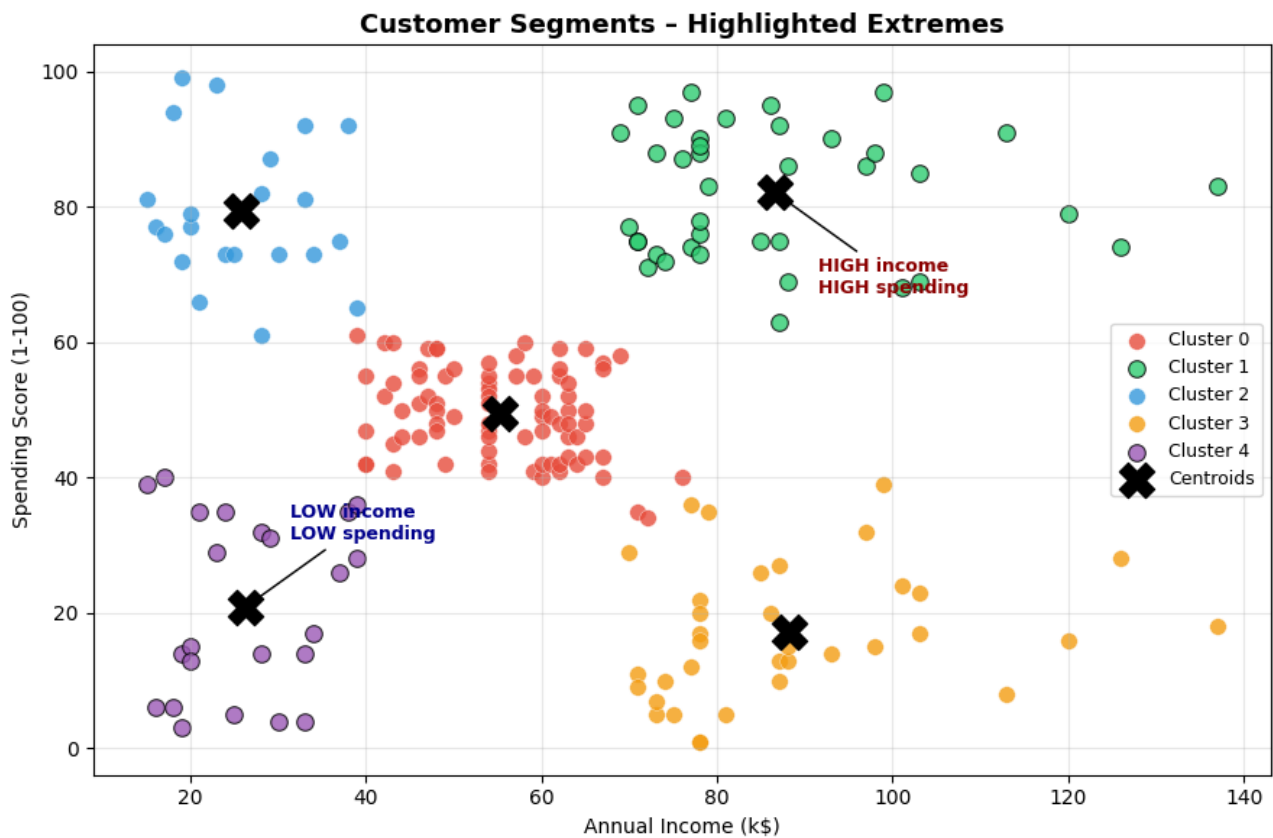


Q. Apply K-Means clustering on the Mall Customers dataset to segment customers based on purchasing behavior and visualize the clusters with centroids.

ANS:



Customer segments:

Cluster	Avg Income (k\$)	Avg Spending Score	Segment Description
0	~55	~50	Moderate income, moderate spending – average customers
1	~26	~79	Low income, high spending – impulsive / lifestyle spenders
2	~86	~82	High income, high spending – prime / premium customers
3	~88	~17	High income, low spending – conservative savers
4	~26	~21	Low income, low spending – budget-conscious customers